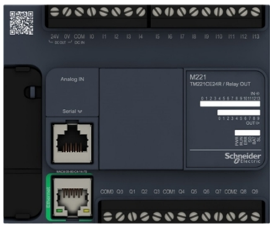
Project Name: New project

CONTENTS

BILL OF MATERIAL

Controller

	**Reference**	TM221CE24R
	Description	TM221CE24R (screw) 14 digital inputs, 10 relay outputs (2 A), 2 analog inputs, 1 serial line port, 1 Ethernet port, 100-240 Vac power supply controller with removable terminal blocks.
	Power supplied to the IO bus	5V: 520 mA / 24V: 160 mA

Cartridge

	**Reference**	TMC2AI2
	Description	TMC2AI2 TMC2 cartridge with 2 analog voltage or current inputs (0 - 10V, 0-20mA, 4 - 20mA), 12 bits.

HARDWARE CONFIGURATION

MyController - TM221CE24R

Digital Inputs

Used	Address	Filtering	Latch	Run/Stop	Events	Priority	Subroutine
X	%I0.0	3 ms			Not Used		
X	%I0.1	3 ms			Not Used		
X	%I0.2	3 ms			Not Used		
X	%I0.3	3 ms			Not Used		
X	%I0.4	3 ms			Not Used		
X	%I0.5	3 ms			Not Used		
X	%I0.6	3 ms			Not Used		
X	%I0.7	3 ms			Not Used		
	%I0.8	3 ms			Not Used		
	%I0.9	3 ms			Not Used		
	%I0.10	3 ms			Not Used		
	%I0.11	3 ms			Not Used		
	%I0.12	3 ms			Not Used		
	%I0.13	3 ms			Not Used		

Digital Outputs

Used	Address	Status Alarm	Fallback value	Used by
X	%Q0.0		0	User logic
X	%Q0.1		0	User logic
X	%Q0.2		0	User logic
X	%Q0.3		0	User logic
X	%Q0.4		0	User logic
X	%Q0.5		0	User logic
X	%Q0.6		0	User logic
X	%Q0.7		0	User logic
X	%Q0.8		0	User logic
X	%Q0.9		0	User logic

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
	%IW0.0	0 - 10 V	Normal	0-1000	0	
	%IW0.1	0 - 10 V	Normal	0-1000	0	

Fast Counters

Used	Address	Input	Configured	Preset	Double Word
	%FC0	%I0.2	NotUsed	0	
	%FC1	%I0.3	NotUsed	0	
	%FC2	%I0.4	NotUsed	0	
	%FC3	%I0.5	NotUsed	0	

High Speed Counters

Used	Address	Type
	%HSC0	Not Configured
	%HSC1	Not Configured
	%HSC2	Not Configured
	%HSC3	Not Configured

ETH1

Device name:	M221
IP Mode:	Fixed
IP address:	0.0.0.0
Subnet mask:	0.0.0.0
Gateway address:	0.0.0.0
Transfer Rate:	Auto
Security Parameters:	Programming protocol disabled
	Auto discovery protocol disabled
	Modbus server disabled
	EtherNet/IP protocol disabled

SL1 (Serial line)

Physical Settings

Device:	None
Baud rate:	19200
Parity:	Even
Data bits:	8
Stop bits:	1
Physical medium:	RS-485
Polarization:	No

Protocol Settings

Protocol:	Modbus
Response timeout (× 100 ms):	10
Time between frames (ms):	10
Transmission mode:	RTU
Addressing:	Slave
Address:	1

Cartridges

TMC2AI2

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
X	%IW0.100	4 - 20 mA	Normal	4000-20000	0	
	%IW0.101	4 - 20 mA	Normal	4000-20000	0	

SOFTWARE CONFIGURATION

Constant Words

KW

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment

KD

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment

KF

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment

Network Objects

Input Assembly (Ethernet/Ip)

Used	Address	Symbol	Fallback value	Comment

Output Assembly (Ethernet/Ip)

Used	Address	Symbol	Comment

Input Registers (Modbus Tcp)

Used	Address	Symbol	Fallback value	Comment

Output Registers (Modbus Tcp)

Used	Address	Symbol	Comment

Digital inputs (IOScanner)

Used	Address	Channel	Symbol	Comment

Digital outputs (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment

Input registers (IOScanner)

Used	Address	Channel	Symbol	Comment

Output registers (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment

Software Objects

Timers

Allocation: Automatic

Allocated: 2

Used	Address	Symbol	Type	Retentive	Dynamic Preset	Time Base	Preset	Comment
X	%TM0		TON			1 s	180	
X	%TM1		TON			1 s	300	DELAY_START

Counters

Allocation: Automatic

Allocated: 0

LIFO/FIFO Registers

Allocation: Automatic

Allocated: 0

Drums

Allocation: Automatic

Allocated: 0

Shift Bit Registers

Allocation: Automatic

Allocated: 0

Step Counters

Allocation: Automatic

Allocated: 0

Schedule Blocks

Allocation: Automatic

Allocated: 0

RTC

PID

Used	PID	Symbol	Type	Comment

Grafcet Steps

Allocation:	Automatic
Allocated:	0

PROGRAM

Behavior

Functional level:	Level 14.0
Starting mode:	Start In Run
Watchdog:	250 ms
Fallback behavior:	Fallback value
String end character:	CR (Carriage Return)

Memory consumption

A successful compilation is required to obtain memory information.

Application Architecture

Master Task

Scan mode:	Normal
POU list:	1 - POU_Inputs
	2 - POU_Scaling
	3 - Main_Program
	4 - POU_Modes
	5 - POU_Valve_Logic
	6 - POU_Outputs

Periodic Task

Period:	255 ms

POU

Master Task

1 - POU_Inputs

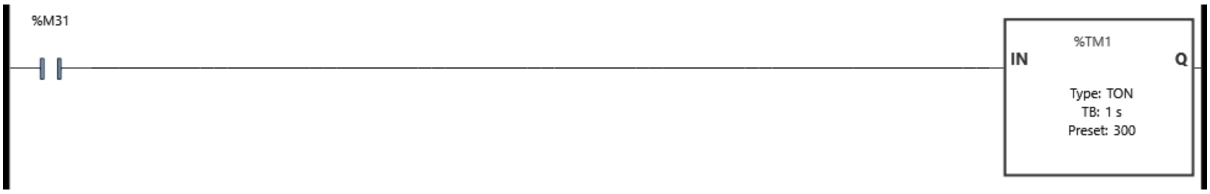
Master Task

Rung0

Variables used:

%M31	POWER_DELAY_ACTIVE	
%S1	SB_WARMSTART	Indicates there was a warm start with data backup

Rung1



Variables used:

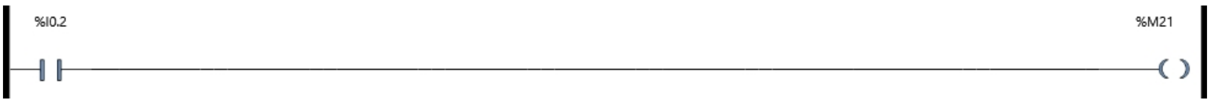
%M31	POWER_DELAY_ACTIVE	
%TM1		DELAY_START

Rung2

Variables used:

%I0.3	DAIKIN_ALARM	
%M4	ALARM_INT	

Rung3



Variables used:

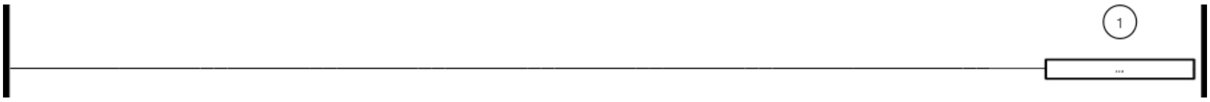
%I0.2	CHILLER_PUMP_FB	pump feedback
%M21	CHILLER_PUMP_FB_INT	

Rung4

Variables used:

%I0.4	DAIKIN_RUN_FB	Fault Signal (80-81)
%M30	DAIKIN_RUN_FB_INT	

Rung5



Legend:

```
1    %MW2 := %IW0.100
```

Variables used:

%IW0.100	
%MW2	TEMP_SENSOR_RAW

Rung6

Variables used:

%I0.5	DAIKIN_MODE	Operation Feedback(82-83)
%M5	DAIKIN_MODE_FB	

Rung7

Variables used:

%I0.6	CIRC_PUMP_FB	Water Flow Signal(70-71
%M27	PUMP_FB_INT	

Rung8

Variables used:

%I0.7	DI_VALVE_FB
%M17	VALVE_AT_CHILLER

Rung9

Variables used:

%I0.7	DI_VALVE_FB
%M18	VALVE_AT_BOILER

2 - POU_Scaling

Master Task

Rung0

Legend:

```
1    %MF0 := ((INT_TO_REAL( %MW2 ) - 4000.0) / 16000.0) * 80.0 - 20.0
```

Variables used:

```
%MW2                TEMP_SENSOR_RAW
```

3 - Main_Program

Master Task

Rung0

Legend:

1 %TM1.Q

Variables used:

%I0.0	MAIN_START	System Start Button
%I0.1	MAIN_STOP	Emergency Stop Button
%M0	SYSTEM_READY	
%M4	ALARM_INT	
%TM1.Q		DELAY_START

Rung1

Legend:

1 %TM1.Q

Variables used:

%M31	POWER_DELAY_ACTIVE	
%TM1.Q		DELAY_START

Rung2

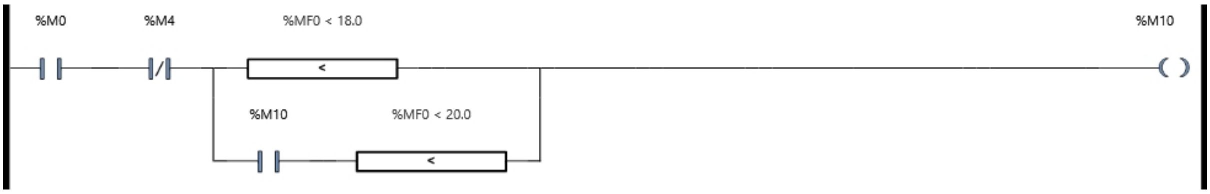
Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M28	CIRC_PUMP_INT_CMD	

4 - POU_Modes

Master Task

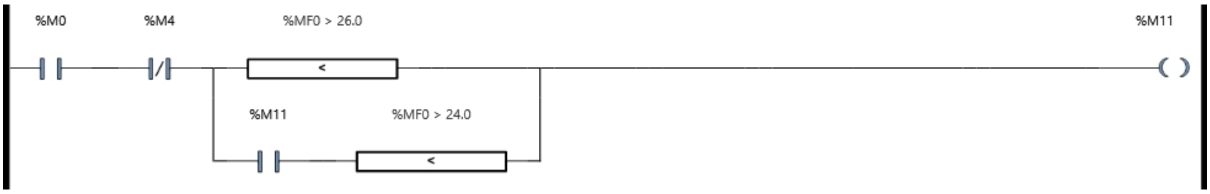
Rung0



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp(REAL)

Rung1



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M11	COOLING_ACTIVE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp(REAL)

Rung2

Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M15	BOILER_MODE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp(REAL)

Rung3

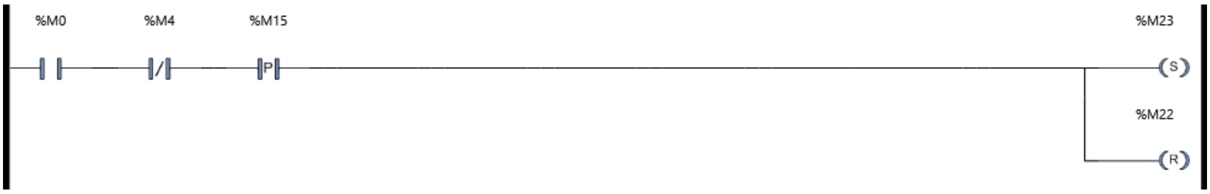
Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M16	CHILLER_MODE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp(REAL)

5 - POU_Valve_Logic

Master Task

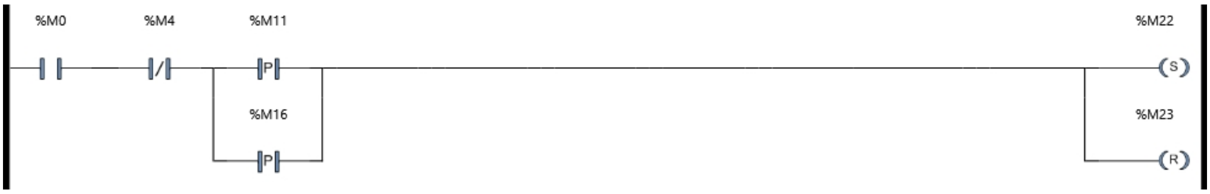
Rung0



Variables used:

%M0	SYSTEM_READY
%M4	ALARM_INT
%M15	BOILER_MODE
%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER

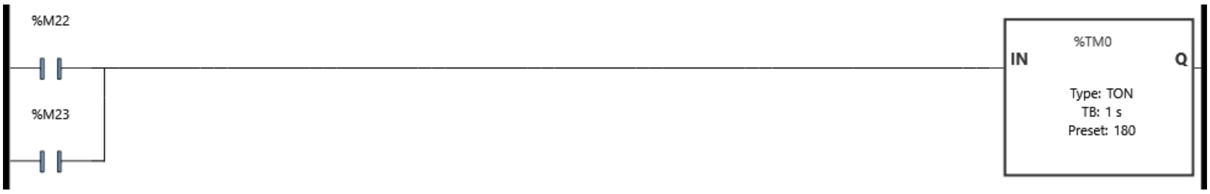
Rung1



Variables used:

%M0	SYSTEM_READY
%M4	ALARM_INT
%M11	COOLING_ACTIVE
%M16	CHILLER_MODE
%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER

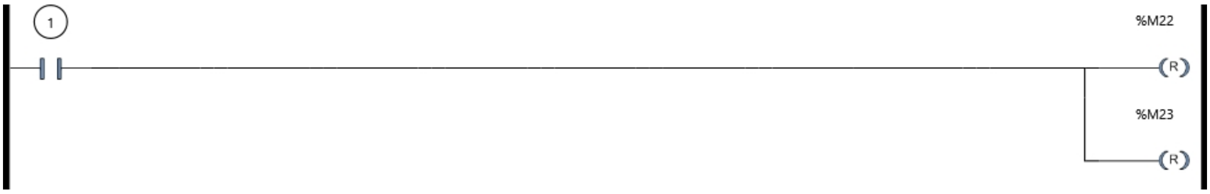
Rung2



Variables used:

%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER
%TM0	

Rung3



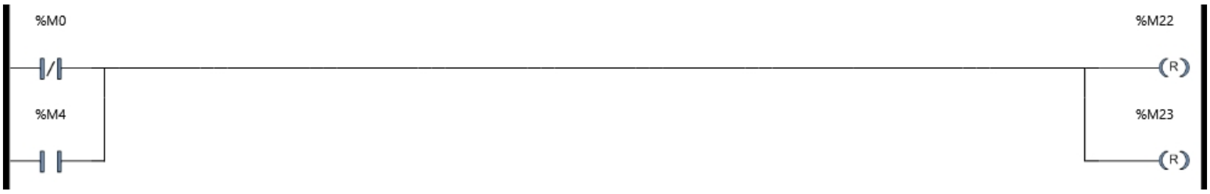
Legend:

1 %TM0.Q

Variables used:

%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER
%TM0.Q	

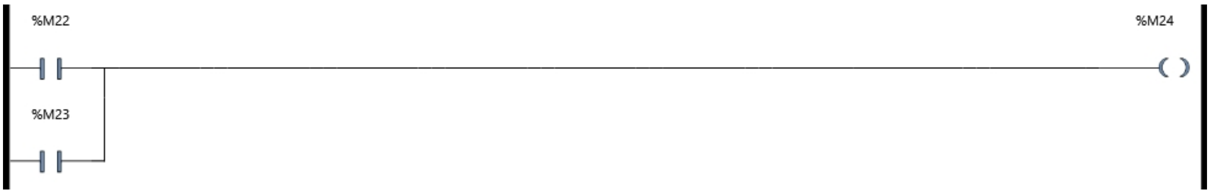
Rung4



Variables used:

%M0	SYSTEM_READY
%M4	ALARM_INT
%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER

Rung5



Variables used:

%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER
%M24	VALVE_TIMER_ACTIVE

6 - POU_Outputs

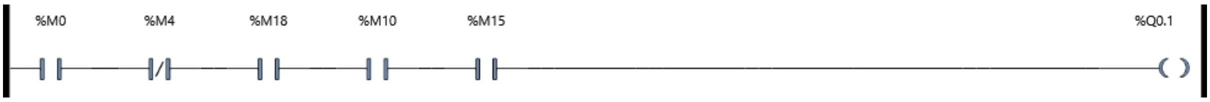
Master Task

Rung0

Variables used:

%M28	CIRC_PUMP_INT_CMD	
%Q0.0	PUMP_START	Pump Control

Rung1



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M15	BOILER_MODE	
%M18	VALVE_AT_BOILER	
%Q0.1	BOILER_START	Daikin Remote ON/OFF(76-77)

Rung2

Variables used:

%M23	VALVE_TO_BOILER	
%Q0.2	VALVE_BOILER	Cool/Heat Selection(72-73)

Rung3

Variables used:

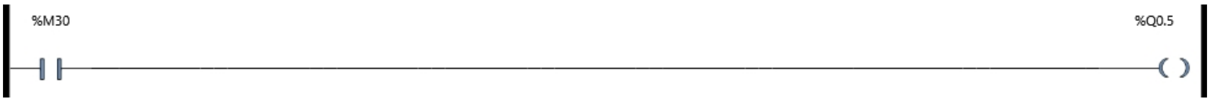
%M22	VALVE_TO_CHILLER	
%Q0.3	VALVE_CHILLER	Boiler Start

Rung4

Variables used:

%M0	SYSTEM_READY	
%Q0.4	LAMP_START	Move to Chiller Position

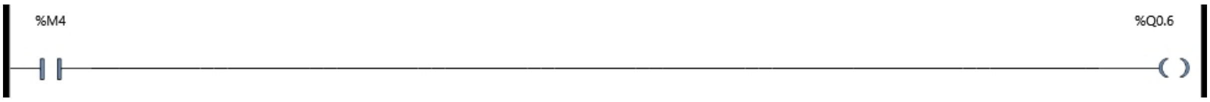
Rung5



Variables used:

%M30	DAIKIN_RUN_FB_INT	
%Q0.5	DAIKIN_RUN	Move to Boiler Position

Rung6



Variables used:

%M4	ALARM_INT	
%Q0.6	LAMP_ALARM	System Alarm Indicator

Rung7

Variables used:

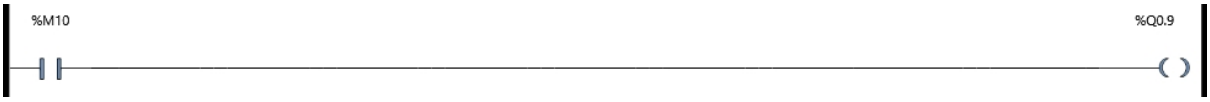
%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M5	DAIKIN_MODE_FB	
%Q0.7	LAMP_RUN	System Operation Indicator

Rung8

Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M11	COOLING_ACTIVE	
%M16	CHILLER_MODE	
%M17	VALVE_AT_CHILLER	
%Q0.8	START_CHILLER	Heating Mode Active Indicator

Rung9



Variables used:

%M10	HEATING_ACTIVE	
%Q0.9	COOLING_HEATING	Cooling Mode Active Indicator

SYMBOLS

Used	Address	Symbol	Comment
X	%I0.0	MAIN_START	System Start Button
X	%I0.1	MAIN_STOP	Emergency Stop Button
X	%I0.2	CHILLER_PUMP_FB	pump feedback
X	%I0.3	DAIKIN_ALARM	
X	%I0.4	DAIKIN_RUN_FB	Fault Signal (80-81)
X	%I0.5	DAIKIN_MODE	Operation Feedback(82-83)
X	%I0.6	CIRC_PUMP_FB	Water Flow Signal(70-71
X	%I0.7	DI_VALVE_FB	
X	%M0	SYSTEM_READY	
X	%M4	ALARM_INT	
X	%M5	DAIKIN_MODE_FB	
X	%M10	HEATING_ACTIVE	
X	%M11	COOLING_ACTIVE	
X	%M15	BOILER_MODE	
X	%M16	CHILLER_MODE	
X	%M17	VALVE_AT_CHILLER	
X	%M18	VALVE_AT_BOILER	
X	%M21	CHILLER_PUMP_FB_INT	
X	%M22	VALVE_TO_CHILLER	
X	%M23	VALVE_TO_BOILER	
X	%M24	VALVE_TIMER_ACTIVE	
X	%M27	PUMP_FB_INT	
X	%M28	CIRC_PUMP_INT_CMD	
X	%M30	DAIKIN_RUN_FB_INT	
X	%M31	POWER_DELAY_ACTIVE	
X	%MF0	ACTUAL_TEMP	Processed Outdoor Temp(REAL)
X	%MW2	TEMP_SENSOR_RAW	
X	%Q0.0	PUMP_START	Pump Control
X	%Q0.1	BOILER_START	Daikin Remote ON/OFF(76-77)
X	%Q0.2	VALVE_BOILER	Cool/Heat Selection(72-73)
X	%Q0.3	VALVE_CHILLER	Boiler Start

Used	Address	Symbol	Comment
X	%Q0.4	LAMP_START	Move to Chiller Position
X	%Q0.5	DAIKIN_RUN	Move to Boiler Position
X	%Q0.6	LAMP_ALARM	System Alarm Indicator
X	%Q0.7	LAMP_RUN	System Operation Indicator
X	%Q0.8	START_CHILLER	Heating Mode Active Indicator
X	%Q0.9	COOLING_HEATING	Cooling Mode Active Indicator
X	%S1	SB_WARMSTART	Indicates there was a warm start with data backup

CROSS-REFERENCE TABLE

Address	Object	Rung	Code
%I0.0....	3 - Main_Program	Rung0	--|P|--
%I0.1....	3 - Main_Program	Rung0	--| |--
%I0.2....	1 - POU_Inputs	Rung3	--| |--
%I0.3....	1 - POU_Inputs	Rung2	--| |--
%I0.4....	1 - POU_Inputs	Rung4	--| |--
%I0.5....	1 - POU_Inputs	Rung6	--| |--
%I0.6....	1 - POU_Inputs	Rung7	--| |--
%I0.7....	1 - POU_Inputs	Rung8	--| |--
		Rung9	--|/|--
%IW0.100.	1 - POU_Inputs	Rung5	--[...]-- %MW2 := %IW0.100
%M0......	3 - Main_Program	Rung0	--( )--
			--| |--
		Rung2	--| |--
	4 - POU_Modes	Rung0	--| |--
		Rung1	--| |--
		Rung2	--| |--
		Rung3	--| |--
	5 - POU_Valve_Logic	Rung0	--| |--
		Rung1	--| |--
		Rung4	--|/|--
	6 - POU_Outputs	Rung1	--| |--
		Rung4	--| |--
		Rung7	--| |--
		Rung8	--| |--
%M4......	1 - POU_Inputs	Rung2	--( )--
	3 - Main_Program	Rung0	--|/|--
		Rung2	--|/|--
	4 - POU_Modes	Rung0	--|/|--
		Rung1	--|/|--
		Rung2	--|/|--
		Rung3	--|/|--

<table>
<tr><th>Address</th><th>Object</th><th>Rung</th><th>Code</th></tr>
<tr><td></td><td>5 - POU_Valve_Logic</td><td>Rung0</td><td>--|/|--</td></tr>
<tr><td></td><td></td><td>Rung1</td><td>--|/|--</td></tr>
<tr><td></td><td></td><td>Rung4</td><td>--| |--</td></tr>
<tr><td></td><td>6 - POU_Outputs</td><td>Rung1</td><td>--|/|--</td></tr>
<tr><td></td><td></td><td>Rung6</td><td>--| |--</td></tr>
<tr><td></td><td></td><td>Rung7</td><td>--|/|--</td></tr>
<tr><td></td><td></td><td>Rung8</td><td>--|/|--</td></tr>
<tr><td>%M5......</td><td>1 - POU_Inputs</td><td>Rung6</td><td>--( )--</td></tr>
<tr><td></td><td>6 - POU_Outputs</td><td>Rung7</td><td>--| |--</td></tr>
<tr><td>%M10.....</td><td>4 - POU_Modes</td><td>Rung0</td><td>--( )--</td></tr>
<tr><td></td><td></td><td></td><td>--| |--</td></tr>
<tr><td></td><td></td><td>Rung2</td><td>--| |--</td></tr>
<tr><td></td><td></td><td>Rung3</td><td>--| |--</td></tr>
<tr><td></td><td>6 - POU_Outputs</td><td>Rung1</td><td>--| |--</td></tr>
<tr><td></td><td></td><td>Rung9</td><td>--| |--</td></tr>
<tr><td>%M11.....</td><td>4 - POU_Modes</td><td>Rung1</td><td>--( )--</td></tr>
<tr><td></td><td></td><td></td><td>--| |--</td></tr>
<tr><td></td><td>5 - POU_Valve_Logic</td><td>Rung1</td><td>--|P|--</td></tr>
<tr><td></td><td>6 - POU_Outputs</td><td>Rung8</td><td>--| |--</td></tr>
<tr><td>%M15.....</td><td>4 - POU_Modes</td><td>Rung2</td><td>--( )--</td></tr>
<tr><td></td><td>5 - POU_Valve_Logic</td><td>Rung0</td><td>--|P|--</td></tr>
<tr><td></td><td>6 - POU_Outputs</td><td>Rung1</td><td>--| |--</td></tr>
<tr><td>%M16.....</td><td>4 - POU_Modes</td><td>Rung3</td><td>--( )--</td></tr>
<tr><td></td><td>5 - POU_Valve_Logic</td><td>Rung1</td><td>--|P|--</td></tr>
<tr><td></td><td>6 - POU_Outputs</td><td>Rung8</td><td>--| |--</td></tr>
<tr><td>%M17.....</td><td>1 - POU_Inputs</td><td>Rung8</td><td>--( )--</td></tr>
<tr><td></td><td>6 - POU_Outputs</td><td>Rung8</td><td>--| |--</td></tr>
<tr><td>%M18.....</td><td>1 - POU_Inputs</td><td>Rung9</td><td>--( )--</td></tr>
<tr><td></td><td>6 - POU_Outputs</td><td>Rung1</td><td>--| |--</td></tr>
<tr><td>%M21.....</td><td>1 - POU_Inputs</td><td>Rung3</td><td>--( )--</td></tr>
<tr><td>%M22.....</td><td>5 - POU_Valve_Logic</td><td>Rung0</td><td>--(R)--</td></tr>
<tr><td></td><td></td><td>Rung1</td><td>--(S)--</td></tr>
<tr><td></td><td></td><td>Rung2</td><td>--| |--</td></tr>
<tr><td></td><td></td><td>Rung3</td><td>--(R)--</td></tr>
</table>

Address	Object	Rung	Code
		Rung4	--(R)--
		Rung5	--| |--
	6 - POU_Outputs	Rung3	--| |--
%M23.....	5 - POU_Valve_Logic	Rung0	--(S)--
		Rung1	--(R)--
		Rung2	--| |--
		Rung3	--(R)--
		Rung4	--(R)--
		Rung5	--| |--
	6 - POU_Outputs	Rung2	--| |--
%M24.....	5 - POU_Valve_Logic	Rung5	--( )--
%M27.....	1 - POU_Inputs	Rung7	--( )--
%M28.....	3 - Main_Program	Rung2	--( )--
	6 - POU_Outputs	Rung0	--| |--
%M30.....	1 - POU_Inputs	Rung4	--( )--
	6 - POU_Outputs	Rung5	--| |--
%M31.....	1 - POU_Inputs	Rung0	--(S)--
		Rung1	--| |--
	3 - Main_Program	Rung1	--(R)--
%MF0.....	2 - POU_Scaling	Rung0	--[...]-- %MF0 := ((INT_TO_REAL (%MW2) - 4000.0) / 16000.0) * 80.0 - 20.0
	4 - POU_Modes	Rung0	--[<]-- %MF0 < 18.0
			--[<]-- %MF0 < 20.0
		Rung1	--[<]-- %MF0 > 26.0
			--[<]-- %MF0 > 24.0
		Rung2	--[<]-- %MF0 < -15.0
		Rung3	--[<]-- %MF0 > -12.0
%MW2.....	1 - POU_Inputs	Rung5	--[...]-- %MW2 := %IW0.100
	2 - POU_Scaling	Rung0	--[...]-- %MF0 := ((INT_TO_REAL (%MW2) - 4000.0) / 16000.0) * 80.0 - 20.0
%Q0.0....	6 - POU_Outputs	Rung0	--( )--
%Q0.1....	6 - POU_Outputs	Rung1	--( )--
%Q0.2....	6 - POU_Outputs	Rung2	--( )--
%Q0.3....	6 - POU_Outputs	Rung3	--( )--

Address	Object	Rung	Code
%Q0.4....	6 - POU_Outputs	Rung4	--( )--
%Q0.5....	6 - POU_Outputs	Rung5	--( )--
%Q0.6....	6 - POU_Outputs	Rung6	--( )--
%Q0.7....	6 - POU_Outputs	Rung7	--( )--
%Q0.8....	6 - POU_Outputs	Rung8	--( )--
%Q0.9....	6 - POU_Outputs	Rung9	--( )--
%S1......	1 - POU_Inputs	Rung0	--| |--
%TM0.....	5 - POU_Valve_Logic	Rung2	%TM0
%TM0.Q...	5 - POU_Valve_Logic	Rung3	--| |--
%TM1.....	1 - POU_Inputs	Rung1	%TM1
%TM1.Q...	3 - Main_Program	Rung0	--| |--
		Rung1	--| |--